

**ALGORITHMS AND COMPLEXITY
PROJECT PROPOSAL:
THE GEOMETRY OF AI DECISIONS**

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BRIEF DISCUSSION ABOUT TOPIC

The core of this project is to implement a k-Nearest Neighbors (k-NN) machine learning classifier from scratch, entirely avoiding external libraries like Scikit-Learn. The k-NN algorithm, originally introduced by Cover and Hart [1], relies on spatial geometry, classifying an unknown data point based on the majority vote of its closest neighbors in a vector space [2]. Our research focuses on how altering the fundamental geometric formulas used to calculate distance, specifically contrasting Euclidean Distance against Manhattan Distance, shifts the algorithm's boundary decisions, judgment, and overall accuracy.

Euclidean distance measures the straight-line distance ("as the crow flies"), while Manhattan distance measures distance along axes at right angles ("taxi driving through city blocks") [3]. We will evaluate the execution dynamics and mathematical tradeoffs of these metrics across a custom 2D dataset, analyzing the algorithm's time complexity and addressing architectural edge cases such as exact equidistant ties or split votes [4].

INPUT AND OUTCOME OF PROPOSAL

- **Inputs:**
 - A custom-curated, non-linearly separable 2D mock dataset featuring items classified by two distinct numerical features, such as Weight vs. Sweetness to classify a fruit category.
 - Coordinate arrays representing strategic test data points positioned directly on or near the decision boundary lines.
 - Configurable hyperparameter variables, specifically the value of k (number of neighbors) and the selection toggle for the distance metric formula (Euclidean vs. Manhattan) [5].
- **Outcomes:**
 - A functional, zero-dependency 2D k-NN classification engine built completely from scratch.
 - A visual 2D scatter plot mapping the dataset coordinates, decision boundaries, and test point classifications [2].
 - A comparative empirical analysis documenting shifting classification results, boundary sensitivity, error rates, and algorithmic execution profiles when swapping geometric metrics across varying k values [3].

REFERENCES

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